

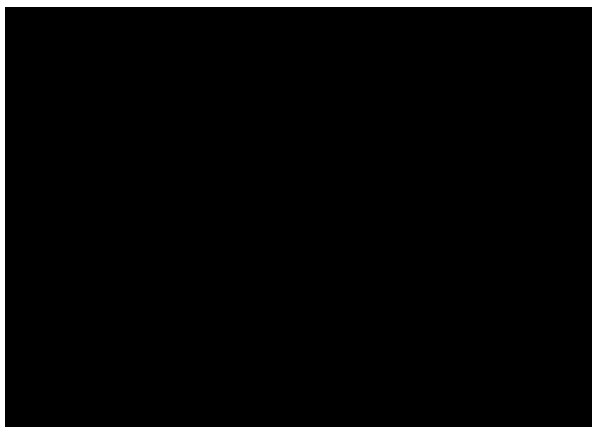
Individual Pathways

Activity 6.2

Investigating the atom

1. Insert a word to complete each sentence.
 - (a) John Dalton proposed that all matter consisted of tiny particles called.....
 - (b) J.J.Thomson proposed that atoms consisted of a sphere containing negatively charged electrons.
 - (c) Rutherford's alpha particle experiments revealed that atoms contained a central positive
 - (d) The nucleus of the atom contains protons and.....
 - (e) Bohr proposed that were located in fixed energy shells around the nucleus.

2. The following apparatus was used by Rutherford to investigate the structure of the atom.



- (a) Identify which of the following elements could be a source (X) of radiation: hydrogen, lead, gold, radium, carbon.
.....

- (b) X is protected by a box Y. Which of the following is a suitable material for box Y: carbon, gold, lead, copper.
.....

- (c) Identify the radiation labelled Z:

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- (d) Most of the radiation strikes the screen without deflection. What does this imply about the atom?

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- (e) Explain why some radiation is deflected backwards at great angles.

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3. Complete the following table. You may need to refer to the periodic table.

Element	Symbol	Number of protons	Number of neutrons	Number of electrons
Fluorine	${}^{19}_{9}\text{F}$			
Potassium		19	21	
	${}^{239}_{94}\text{Pu}$			
Krypton		36	56	
	${}^{141}_{56}\text{Ba}$			

4. The model of the atom produced by Niels Bohr had electrons arranged in energy levels or shells at different radial distances from the nucleus. The first shell ($n=1$; closest to the nucleus) was called the K shell. The table shows the shell number and name. Given that the maximum number of electrons in each shell 'n' is $2n^2$, complete the table. The first row has been completed for you.

Shell number (n)	Shell letter	Maximum electron number
1	K	$2(1)^2 = 2$
2	L	
3	M	
4	N	
5	O	

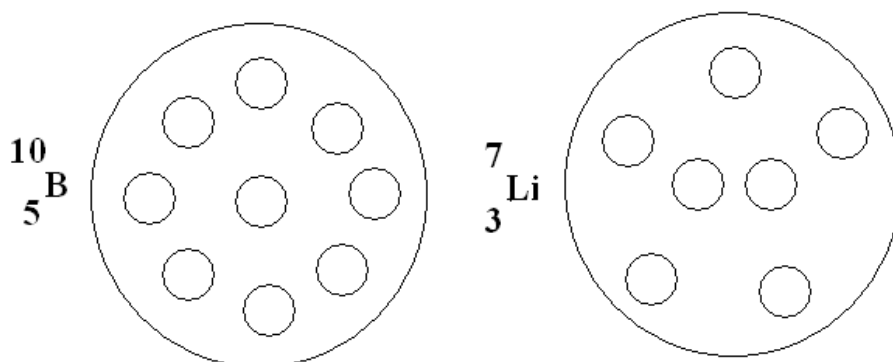
5. Each chemical element has a different number of protons and electrons. Electrons fill the electron shells starting from the K shell and working outwards. Once a shell is filled, any remaining electrons start to fill the next shell. Thus beryllium (Be) has four electrons and the first two electrons fill the K shell. The remaining two electrons are placed in the L shell.

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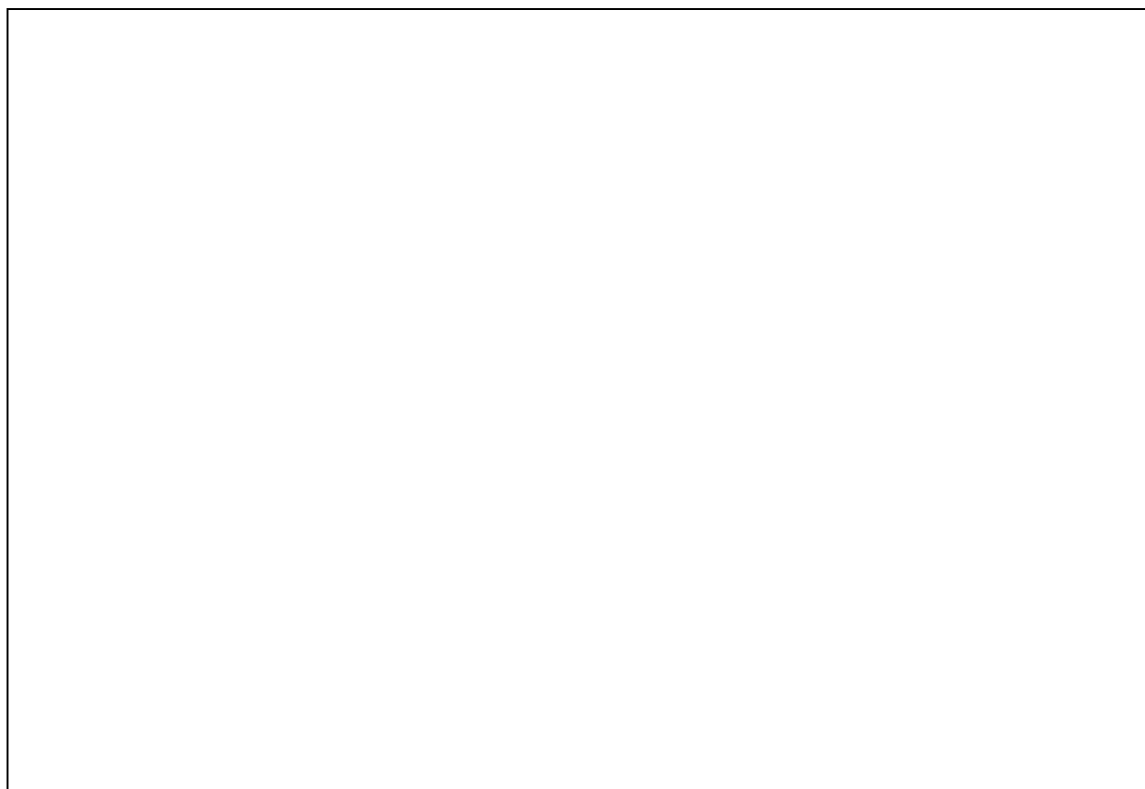
Based on this information, complete the following table.

Element	Z	K shell	L shell	M shell
Carbon	6			
Sodium	11			
Silicon	14			
Chlorine	17			

6. The following diagram shows the nuclei of various atoms. Each circle represents a proton or a neutron. In each example, shade in the neutrons that are present.

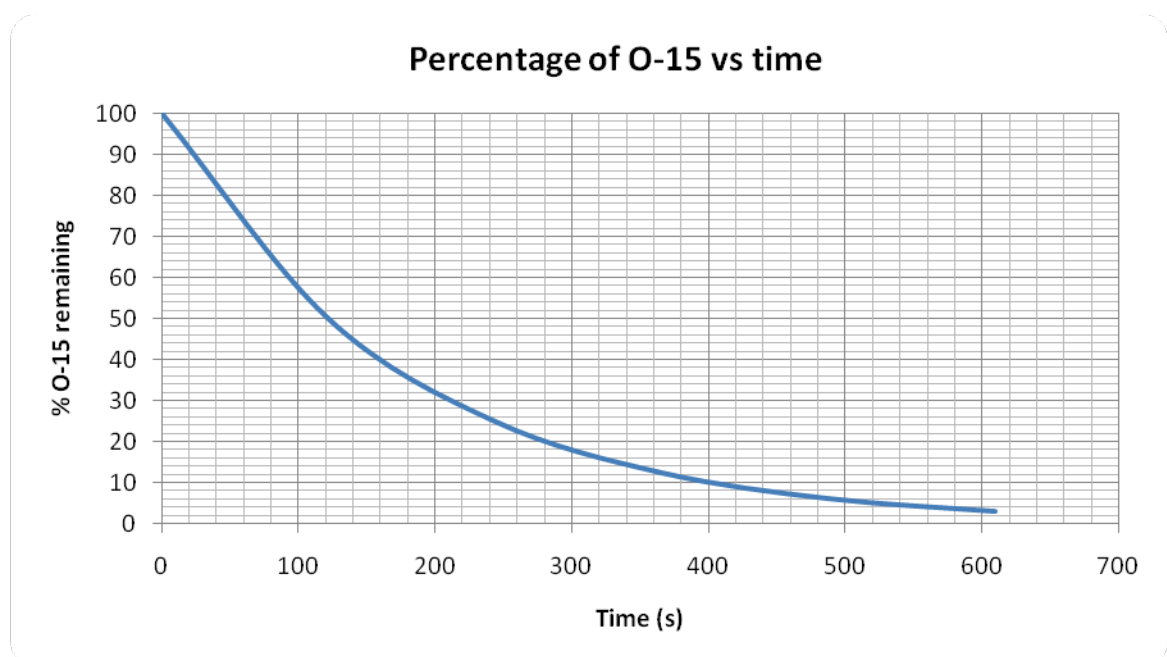


7. Lead-211 has a half-life of 36 minutes. A 50 g sample of Pb-211 was obtained and the mass remaining measured as a function of time. Construct a line graph of the mass of lead-211 remaining after six half-lives have elapsed.



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8. The following graph shows the decay curve for oxygen-15.



Using the graph, estimate the half-life of oxygen-15.

9. Bismuth-210 ($Z=83$) radioactively decays to form polonium-210 ($Z=84$).

(a) Calculate the number of protons and neutrons in the bismuth-210 isotope.

.....

(b) Calculate the number of protons and neutrons in the polonium-210 isotope.

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(c) Explain what has happened during this nuclear decay.

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10. In the natural decay of U-238 ($Z=92$), the daughter nucleus formed is Th-234 ($Z=90$). Determine the type of radiation emitted.

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